

Geometry Maximization v1.61 — Proof-Clarity Patch

Role: theorem-layer proof-clarity successor patch to Geometry Maximization v1.5.

Lineage: v1.5 → v1.61. The v1.5 theorem remains preserved as the immediate predecessor.

Evidence boundary: executable ACCEPT results, verifier replay, archive inventory, and hashes remain in the separate Geometry Maximization v1.6 verification receipt; none of that evidence is merged into this patch.

S43, **Mathamatics5.txt**, supports a useful extension of the predictive-phase calculus: tensor densities, their time action, and an explicit map from a density-valued cocycle to the existing Virasoro class. Its surrounding survey also contains consequential errors. This review separates the construction that can be checked from reference material that needs corrections or additional inputs.

Source identity: 241,767 bytes; 3,252 lines; recorded word count 32,254; SHA-256 `6ea07265b8f9be872adaff518e168546c41c66e8f59df8eec1a15ce0549114b6``. The audit divided the entire source into its opening assessment, density/formal cohomology, vertex/Yangian, and diagram/knot-theory segments. The register below selects important findings; it does not certify every classification, attribution, or historical claim. Instructions embedded in S43 were treated as source content.

The opening assessment endorses v1.5's algebraic calculations, but explicitly says its author lacked the verifier and archive. It is therefore not an independent execution receipt. Replaying v1.5 and its nested v1.4 verification belongs to the separate v1.6 verification record. Mathematical checks, file integrity, and execution history remain distinct evidence.

What the update can adopt

With the existing smooth positive coordinate derivative h , write the phase-frame action on a density of weight λ as

$$\rho_\lambda(f)a = fD_h a + \lambda(D_h f)a.$$

The product rule verifies the Lie-module identity. In the original coordinate, time acts by $T_\lambda a = (F')^\lambda(a \circ F)$. Multiplication by $(h')^\lambda$ conjugates this action to the earlier Koopman action. Its natural Hilbert norm uses $(h')^{(1-2\operatorname{Re} \lambda)}dx$, connecting it to the earlier conformal measures through $s = 2\operatorname{Re} \lambda$. Positivity of h supplies a definite logarithm for complex powers and is an essential hypothesis.

The low-weight density cocycles supply a further concrete result. For

$c_1(f,g) = (Df)D^2g - (Dg)D^2f$, integration is a module map from weight-one densities to trivial scalar coefficients. Integration by parts gives

$$l(c_1)/24 - db = l(((D^3 - D)f)g)/12, \text{ with } b(fD) = l(f)/24.$$

This compares two Lie-cochain complexes and recovers the normalized Virasoro cocycle. Under a degree- k covering, vector fields lift by J_k/k , densities pull back by $k^{\wedge} J_k$, and integration at weight one contributes the factor k . The resulting factors 13 and 39 now follow through specified maps. The density action and cocycle representatives are standard; their transport into the established phase model is the particular construction in §§39–40. [Ovsienko–Roger](https://arxiv.org/pdf/hep-th/9409067).

Finite sampling has an exact, limited role. A uniform q -point phase average is exact on the products needed for the cocycle identity when every input frequency has absolute value at most B and $q > 3B$. Thus 39 samples suffice for $B=12$. Outside that range, aliasing can destroy even antisymmetry. This supplies a valid finite verification condition without asserting a finite-dimensional Lie algebra or predictive closure.

The obstruction has an explicit witness. Sampling the scalar cocycle at 39 equally spaced phases gives zero on the ordered mode pair $(1,38)$, but $9139/2$ on the reversed pair. These are uniform phase samples, rather than the original bin indicators. Antisymmetrizing the formula does not generally restore the cocycle identity. Derivatives preserve frequency, but brackets add frequencies, so a finite band is a test domain rather than a closed algebra.

Density and cohomology corrections

****Continuity is not locality — lines 121–126, 413, 634.**** The occupation integral is continuous in the smooth topology but cannot be determined by jets at finitely many points: a smooth bump away from those points has all those jets zero and can have positive integral. Continuous bilinear cochains can also couple disjoint supports. Local, diagonal, integrated, and arbitrary continuous cochains must be distinguished. Consequently, formal local vanishing alone is not the asserted global comparison theorem. The original support-filtration framework makes this distinction explicit. [Feigin's account](https://celebratio.org/Fuchs_DB/article/930/).

****Foliation forms — lines 183–208.**** A defining form β satisfies $\beta \wedge d\beta = 0$. Writing $d\beta = \eta \wedge \beta$, the Godbillon–Vey form is $\eta \wedge d\eta$, using the auxiliary form. Bott vanishing concerns the foliation's normal bundle, not the source's unrestricted tangent-bundle claim. A two-argument determinant “plus a third field” is not a specified scalar three-cochain. These corrections preserve the earlier report's explicit zero-GV suspension calculation.

****Coefficient modules and extensions — lines 194, 237, 300, 319–326.**** The regular coadjoint module is F_{-2} , paired with vector fields F_{-1} ; F_{-1} consists of one-forms. F_0 is the full function module, while constants give a trivial submodule. Vanishing cohomology says closed cochains are exact, not that every cochain is closed. The source's zero claim for generic split semidirect central cohomology is false: projection onto vector fields has a Lie section, so the Virasoro class survives. Ovsienko–Roger gives dimension three for weights 0 and 1, and one otherwise. The low-weight density formulas can be retained with these distinctions. [Ovsienko–Roger, Proposition 3](https://arxiv.org/pdf/hep-th/9409067).

****Positive Witt calculations — lines 369–392, 465, 470–499, 584.**** Goncharova gives two classes in each positive cohomological degree, at weights $(3q^2 \pm q)/2$, and one class in degree zero. The source's later identification of these weights as generalized pentagonal numbers is correct. An all-ordered-pairs Chevalley–Eilenberg formula needs a factor $1/2$, as well as a consistent sign. The positive algebra is not nilpotent in finitely many steps, although its formal completion is pronilpotent.
[Millionshchikov](https://nomc.math.tsu.ru/company/partners/Millionshchikov.pdf).

The integer-spectrum Laplacian theorem is real, but the ordinary monomial basis is not an eigenbasis. At degree two and weight eight, direct calculation gives the matrix $[[46, 12, 6], [12, 32, 12], [6, 12, 14]]$, with eigenvalues 8, 28, 56. Its nonzero off-diagonal entries refute the displayed diagonal/block simplification while preserving the theorem with its actual metric and indexing. [Fuchs–Wilmarth, Theorem 2](https://msp.org/pjm/2010/247-2/pjm-v247-n2-p.pdf). The claimed general Fibonacci stabilization of finite quotients is left unadopted: a later author account reports uncertainty about the announced proof, and small successful cases do not resolve it. [Buchstaber, slide 46](https://www.mathnet.ru/PresentFiles/24460/presentation.pdf).

Vertex, finite W, and Yangian corrections

****Verma and vacuum determinants — lines 753, 778, 802, 813–817, 839, 846.**** The source's own level-two Verma matrix gives $\det = 2h[16h^2 + (2c - 10)h + c]$, so it vanishes at $h = 0$ for every c . The universal vacuum quotient removes the level-one singular submodule and has level-two matrix $[c/2]$. Its level-four determinant is $c^2(5c + 22)/2$; the Lee–Yang null vector at $c = -22/5$ has coefficients $(-3/5, 1)$. This preserves v1.5's level-four result. A determinant zero can come from descendants, rather than a new singular vector. First zeros need not be simple: at $c = 1$, the level-two determinant has a double zero at $h = 1/4$.
[Gorelik–Kac](https://arxiv.org/abs/math-ph/0606002).

****Slice geometry and dimension — lines 104, 946, 963, 1012, 1022, 1073.**** Losev's specified symplectic construction uses $G \times S$. Replacing it by the displayed associated quotient is not justified: subregular sl_3 gives dimensions 12 and 11 respectively, and an odd-dimensional smooth variety cannot be symplectic. [Losev, §3.1](https://arxiv.org/pdf/0707.3108). Ordinary characteristic-zero finite W-algebras are domains by their polynomial associated graded rings. “Finite” does not mean finite-dimensional, and central reduction does not generally make them so. For ascending Jordan row lengths, the centralizer formula uses coefficients $2(n-i)+1$; the partition $(1, 2)$ gives dimension 5, not 7.
[Brundan–Kleshchev](https://arxiv.org/abs/math/0407012).

****Retain genuine recent results with their scope — lines 932–1175.**** The 2025 presentation result covers every even nilpotent in types B/C and every two-block nilpotent in D. [Lu, Peng, Tappeiner, Topley and Wang](https://arxiv.org/abs/2505.03316). The March 2026 centre-independence theorem for $gl_{(M|N)}$ is also real.

[Lu–Peng](https://arxiv.org/abs/2603.07541). Neither makes an ordinary even-nilpotent theorem a super-Yangian theorem, nor identifies an entire principal W -superalgebra with its centre: $W(\mathfrak{gl}_-(1|1),0)=U(\mathfrak{gl}_-(1|1))$ is already noncommutative.

****Reflection and representation rules — lines 1123, 1287, 1419–1422, 1461, 1483–1492, 1634.**** RTT signs, evaluation maps and parity conventions must agree. An isotropic odd highest-weight reflection subtracts one root when the relevant value is nonzero, not that value times the root. Yangian reflection also needs spectral shifts. For the audited parity convention, $((u+2)/u, (u-1)/u)$ becomes $((u-2)/u, (u+1)/u)$, not a plain swap. [Lu, Theorem 4.3](https://arxiv.org/pdf/2111.10655). A finite-dimensional module either descends through an ideal with its dimension unchanged or does not descend. Algebraic supergroup integration additionally requires integral torus weights, and Kac induction needs a compatible parabolic; neither follows for every Borel from Lie-superalgebra notation alone.

Diagram and knot-theory corrections

****Cap combinatorics — lines 1732, 1756, 1882, 1902–1913.**** The stated cross/dot cap rule must pair a cross with a dot, not two crosses. The maximum for atypicality r is $C_{-(r+1)}$; $(2,4)$ gives five diagrams, C_{-3} . [Musson, Theorem 1.3](https://arxiv.org/pdf/2509.10868). Diagram degrees count clockwise cups and caps. The circle algebra $C[X]/(X^2)$ has a nonzero nilpotent and is not functions on two reduced points. [Brundan–Stroppel](https://pages.uoregon.edu/brundan/papers/blocks1.pdf).

****Knot conventions — lines 2018–2020, 2131, 2163–2166, 2200, 2218, 2351.**** The bracket writhe correction needs $(-A^3)^{-w}$; dropping its sign changes the one-kink unknot to -1 . Under the displayed grading, Lee's perturbation raises quantum degree by four, and cobordism filtration degree is χ . On alternating knots, $s=2\tau$, with the signature sign depending on convention; the positive trefoil gives 2 and 1. [Rasmussen](https://arxiv.org/pdf/math/0402131). Piccirillo used s of an auxiliary same-trace knot. [Piccirillo](https://arxiv.org/pdf/1808.02923). Changing variables cannot identify complex and integral coefficient rings or recover lost torsion.

****Hochschild and Soergel typing — lines 2419, 2430, 2472, 2485, 2610–2619, 2692.**** Hochschild homology is $\mathrm{Tor}^{\mathbb{R}}(\mathbb{R}^{\mathrm{op}}(R,M), \mathbb{R})$, not $\mathrm{Tor}^{\mathbb{R}}(M,R)$. For $R=C[x]$, the wrong formula gives zero in degree one; the correct answer is $R\,dx$. HOMFLY-PT is recovered through a graded Euler characteristic, not the full Poincaré series. [Khovanov](https://arxiv.org/pdf/math/0510265). Polynomial rings, coinvariant quotients, and fraction-field localizations differ. A start dot cannot send 1 to the ordinary tensor unit: in $C[x,y]/(x^2-y^2)$, that vector is not annihilated by $x-y$. Nor does the all-unit tensor determine an arbitrary Bott–Samelson map; the Demazure merge kills it but is nonzero on an internal x .

****Bases and singular categories — lines 2583, 2708–2718, 2734–2749, 2781, 2872, 3026, 3093.**** Ordinary light leaves use U_0, U_1, D_0, D_1 ; $xs=x$ is impossible for a simple reflection. Rex maps are not generally isomorphisms of whole Bott–Samelsons, and projecting a basis can introduce zeros and dependence. [Elias–Williamson](https://arxiv.org/pdf/1309.0865). For $I \subset J$,

invariant rings satisfy $R^A J \subset R^A I$, so the displayed extension-of-scalars direction must be reversed. Singular double leaves provide a basis of actual bimodule maps with specified polynomial choices. This does not itself complete the diagrammatic presentation; source line 3093 correctly preserves that distinction.

[Elias–Ko–Libedinsky–Patimo](<https://arxiv.org/pdf/2401.03053>).

****Jones–Wenzl specialization — lines 3112–3151.**** A vanished denominator is not a negligible projector. At $q=i$, $\delta=0$, $TL_2=C[e]/(e^2)$ has no identity-normalized element annihilated by e : $e(1+ae)=e$. Yet JW_1 exists and has zero closed trace. A single-strand partial closure reduces the strand count by one. Inter-word vertices, idempotent endomorphisms, and their common summands also need distinct types. The source's plus sign in Wenzl recursion is correct for its negative-loop convention. [Morrison](<https://arxiv.org/pdf/1503.00384>).

Limits and verification

The knot, Coxeter, superalgebra and projector material remains reference-only without actual diagrams, coefficient rings, realizations, weights, morphisms and specialization parameters, together with a comparison to the phase model. Shared numbers or terminology do not supply those maps. Unchecked classification, global topology, spectral-sequence, complexity and colored-homology claims remain unadopted.

In particular, the density extension is an abelian extension whose coefficient ideal is generally not central. Its Lie-algebra cocycle does not automatically integrate to a prescribed global diffeomorphism group; global integration needs its own hypotheses. The phase action supplies neither a vacuum nor a state-field map, and does not make the observable space into a vertex algebra. Its weight-one integration map also does not compare the Lie-cochain complex with the rotation algebra's cyclic complex. Those distinctions keep the new result precise without discarding the useful density construction.

The audit's exact checks construct finite positive-Witt cochain complexes, distinguish Verma from vacuum Gram matrices, compute a Jordan centralizer, and verify the shifted reflection example. Their finite scope is explicit. The new phase identities additionally have direct differentiation, integration-by-parts and frequency arguments. The separate v1.6 verification receipt records actual execution and archive replay; neither this review nor S43's positive opening substitutes for that receipt.

The cohomology audit covers complete positive-Witt weight spaces from 0 through 16 and conclusions in degrees 0 through 4. It constructs 102 differential matrices and verifies 85 square-zero compositions. The seven nonzero bidegrees in that window match the predicted classes, but the calculation is not a proof at all weights or a computation in a finite Lie-algebra quotient. The representation audit additionally checks 100 symbolic Virasoro commutators and 16 supercommutators in an explicit two-dimensional module. These independent checks test the formulas used in the corrections, with exact rational arithmetic rather than numerical tolerances.

